Utkarsh

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Education

SRM Institute of Science and Technology Aug. 2023 – Present

B.Tech. in Computer Science, Specialization in Computer Networking *CGPA: 9.67*

- **Coursework:** Computer Networks, Operating Systems, Data Structures, DBMS, Algorithms, Object-Oriented Programming

Skill Summary

Languages: C/C++, JavaScript, TypeScript, HTML/CSS, SQL, Python

Frameworks: React, Node.js, Express, Next.js, PySpark

Developer Tools/Platforms: Git, GitHub, Docker, AWS (EC2, S3, CloudFront)

Databases: MySQL, PostgreSQL, MongoDB

Experience

Microsoft May 2026 – July 2026

Security Research Intern *On-site, Hyderabad*

- Developed a graph-based framework to **detect and reconstruct coordinated, multi-organization device-code phishing campaigns** sharing attack infrastructure using Microsoft's global threat graph data.
- Built **campaign visualization and automated story-generation capabilities** that transformed threat graph data into actionable attack narratives.
- Designed a **campaign-persistence schema and framework for longitudinal tracking**, enabling researchers to measure campaign progression and identify intersections across campaigns.

Projects

Deep Packet Inspection Engine | GitHub 2026

Tech used: C++, CMake, PCAP files

- Built a network traffic analysis engine that **reads packets from `.pcap` captures** and inspects their contents to **function as a software firewall**.
- Implemented **multithreaded packet processing** while preserving connection affinity, ensuring packets from the same flow are handled by the same worker for speed and accuracy.
- Added **traffic filtering** that identifies applications or websites, including encrypted HTTPS traffic, and blocks traffic using IP, application, or domain-pattern rules.

DealDrip | Live Demo | GitHub 2025

Tech used: Next.js, Node.js, Cheerio, Bright Data, MongoDB, Nodemailer

- Built a web application that **monitors Amazon product prices and stock availability**, allowing users to track products and receive automated updates.
- Implemented a **scheduled scraping job that collects product details** and stores structured records in MongoDB for up-to-date tracking.
- Developed an **automated notification system that sends email alerts** for price drops and back-in-stock events.

AI Interview Preparation Platform | Live Demo | GitHub 2025

Tech used: Next.js, React, TypeScript, Firebase Auth & Firestore, Vapi AI, Gemini

- Built an AI-powered mock-interview platform where **users practice technical and behavioral interviews** by voice and receive instant feedback reports.
- Integrated the Vapi voice-agent SDK with Gemini to dynamically **generate role-specific interview questions and structured feedback**.
- Implemented **secure Firebase authentication** with server-side session cookies and used Firestore to store users, interviews, and feedback.

Extracurricular

- Achieved **Knight** rating on LeetCode with a peak rating of **1877**; solved 500+ algorithmic problems across coding platforms. LeetCode Profile
- Won **1st Place** at the AI Project Expo 2025, School of Computing, SRM; finished first among 30 shortlisted teams, received a Rs. 3,000 award, and developed an agentic AI mock-interview platform.
- Placed in the **Top 3 of 150+ teams** at SRM Hackathon 8.0 by developing a space-debris tracking system.